

SUJAL KUMAR KANU

B.E. in Computer Science & Engineering (3rd Year Student)

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EDUCATION

BMS Institute of Technology & Management (VTU) – B.E. in Computer Science & Engineering 2023 – Present (3rd Year)

Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Computer Networks, Operating Systems, Discrete Mathematics.

National Infotech Secondary School – Grade XII (NEB, Nepal) Graduated 2024

TECHNICAL SKILLS

Languages: Java, JavaScript (ES6+), C, C++, Python, SQL, HTML5, CSS3
Web & Frameworks: React 18, Node.js, Express.js, FastAPI (Python), TailwindCSS, Ethers.js, Recharts
Databases & Blockchain: PostgreSQL (Triggers, Views), Redis, Solidity 0.8+, Polygon Amoy Testnet, Hardhat
Tools & Graphics: Docker, Docker Compose, Git, GitHub, Linux (Fedora/Ubuntu), Sketchfab Viewer API, MATLAB
Core Domains: Full-Stack Development, Smart Contracts, RESTful APIs, DSA, IoT & Cybersecurity Basics

TECHNICAL PROJECTS

AgriChain – Blockchain Supply Chain & Fraud Detection | Solidity, Polygon, React, Node.js, Docker 2026

- Architected a decentralized supply chain covering 4 role tiers (Farmer, Distributor, Retailer, Consumer) on Polygon Amoy with Solidity smart contracts and Ethers.js.
- Implemented an automated 3-tier fraud engine flagging duplicates, missing transit steps, and price spikes (>500%) while computing dynamic trust scores (0–100).
- Created batch-specific dynamic QR code generation for end-to-end provenance verification, integrated live logistics tracking, and containerized the service using Docker Compose.

Research Collab Hub – AI-Powered Academic Research Platform | FastAPI, PostgreSQL, Redis, React, Claude API 2026

- Developed a modular collaborative research system with automated paper version archiving, BibTeX citation export, and milestone/deadline management.
- Built an integrated LLM pipeline using Anthropic Claude for abstract summarization, venue suitability analysis, and co-author recommendations via Jaccard similarity.
- Designed a relational PostgreSQL schema utilizing triggers for automated revision tracking, audit logs, and instant notifications; backed by Redis caching.

CardioSim 3D – Interactive 3D Heart Anatomy Simulation | JavaScript (ES6+), Sketchfab 3D API, HTML5, CSS3 2026

- Engineered an interactive 3D human heart simulation platform featuring photorealistic rendering, 360° orbital control, and dynamic anatomical callouts for 24+ structures.
- Built a synchronized cardiac engine dynamically adjusting from 30 BPM to 200 BPM paired with pulsing CSS animations and cardiac cycle phase transitions.
- Simulated cardiac pathologies (arterial blockage, arrhythmia, valve regurgitation) and developed an interactive medical quiz assessment engine.

HACKATHONS & COMPETITIONS

CODE RED 3.0 (24-Hour Hackathon) 2026
E-Cell BMSIT&M – Prototyped and pitched a working software solution in 24 hours.

ArtPark CodeForge – Rhapsody 4.0 Mar 2026
IISc Bangalore – Competed among 2,796+ national teams in rapid software prototyping.

CERTIFICATIONS & LANGUAGES

- Introduction to Cyber Security** – Infosys Springboard (2025)
- Statistics & Curve Fitting Onramp** – MathWorks (2025)
- IoT & Robotics Intensive** – NepaTronix (2023)

Languages: English (Professional), Hindi, Nepali, Bhojpuri